

Do Growth and Onset of Symptoms of Uveal Melanomas Indicate Subclinical Metastasis?

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Abstract: The currently popular thesis that metastatic disease from uveal melanoma begins at about the same time that the intraocular tumors become symptomatic, when they not only exhibit accelerated local growth, but also invade blood vessels and begin to disseminate hematogenously, has been examined in the course of a retrospective analysis of 2,627 cases of choroidal melanoma treated by enucleation over a 40-year period. During the four decades studied, the proportion of smaller tumors with less advanced clinical manifestations increased, but there was no parallel increase in survival. These data, along with the results currently being reported by others who have treated uveal melanomas by methods other than enucleation, or who have followed patients with untreated melanomas for prolonged periods, provide no support for the concept that dissemination leading to metastasis begins with the onset of clinical symptoms. [Key words: dissemination, melanoma, metastasis, mortality after enucleation, prolonged course without metastasis.] *Ophthalmology* 91:685-691, 1984

Our interest in growth patterns of uveal melanoma is closely linked to our interest in deciphering the survival curves that so many investigators have published, recording patterns of survival after enucleation.¹⁻⁶ If one assumes, as virtually everyone did until a few years ago, that enucleation can only have a beneficial effect, then the shape of these curves, if extrapolated backward in time, would suggest that uveal melanomas must be highly malignant, rapidly metastasizing tumors carrying a very high mortality before their detection and treatment (Fig 1). For every 100 patients brought to the operating room for enucleation, there should have been many who had already died of metastatic disease, or who were found to have evidence of metastatic disease at the initial exam-

ination. Deaths from unsuspected uveal melanomas and evidence of metastasis at the time of initial diagnosis have been exceptionally unusual as compared with their well known frequencies with cancers of the respiratory, gastrointestinal, and genito-urinary tracts, carcinoma of the breast, and melanomas of the skin; yet, there is a curiously significant abrupt rise in mortality during the first few years after enucleation (Fig 2). The graph in Figure 3 shows how nicely the data points fit a log-normal distribution curve beginning at the time of enucleation. This indicates that dissemination usually begins at about the time of enucleation. Therefore, those who have steadfastly refused to believe that the operation of enucleation in itself could be so important in the pathogenesis of metastasis have sought other explanations for this temporal relationship of deaths to enucleation.

One of the most popular theories has invoked the concept that while uveal melanomas may begin as rather benign, slowly growing, nonmetastasizing tumors that are asymptomatic, eventually a dramatic biologic change takes place (Fig 4). The tumor becomes more highly malignant. It grows more aggressively in the eye, producing complications that provoke a variety of symptoms that

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